The Impact of Credit Information Sharing on Bank-Specific Stock Price Crash Risk: The Role of Information Asymmetry & Bank Regulations

Chanon Teeranutranont

PIER Research Exchange

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Durham University
Business School

Agenda

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Introduction

- What is “Credit Information Sharing”?
  - A process where banks share/exchange information about their borrowers.

Private Credit Bureau

Public Credit Registry

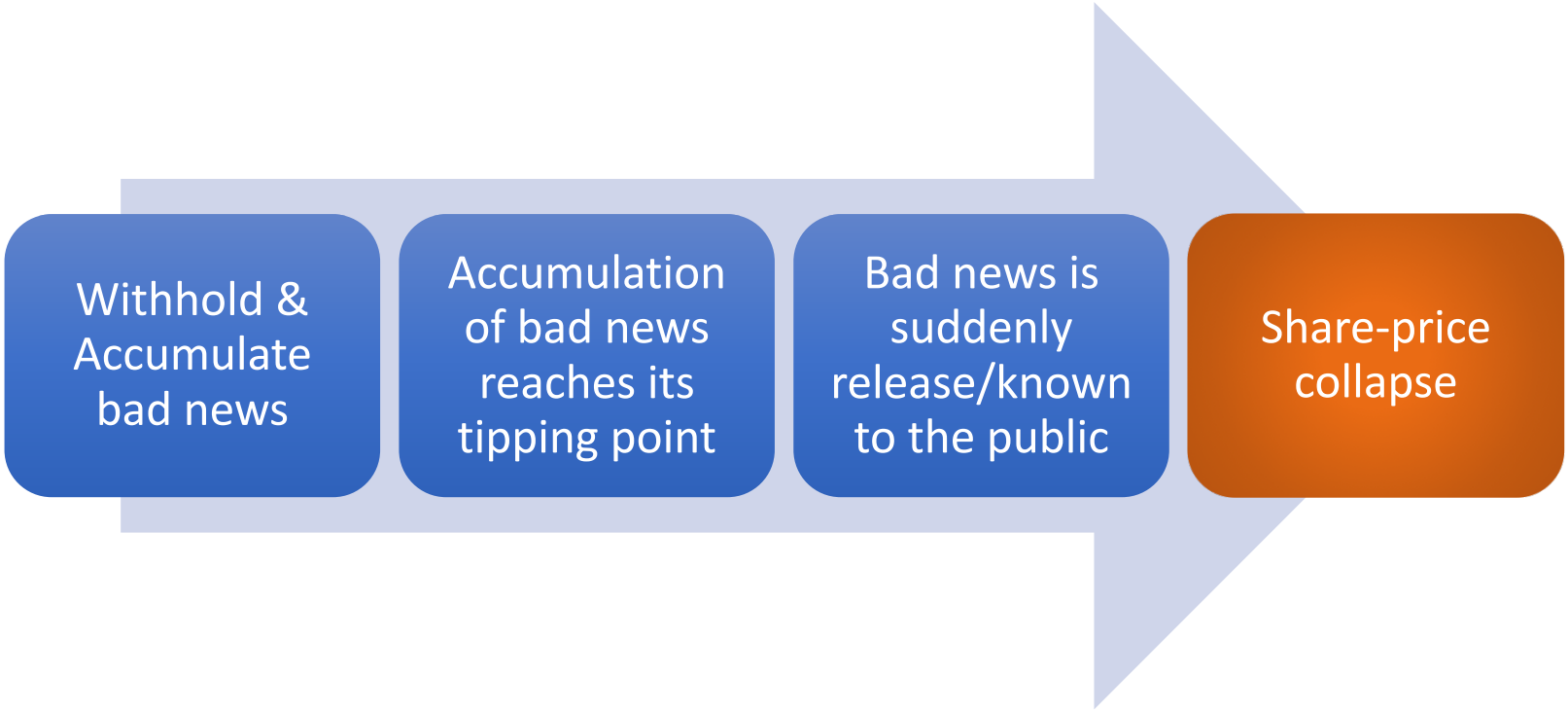
- What is ”Stock Price Crash Risk”?
  - The risk of a large negative stock return

Background

- Several mechanisms could generate stock price crash for instance,…..
  - leverage effect (Black, 1976, Christie, 1982),
  - bubble theories (Blanchard and Watson 1982),
  - investors' difference-of-opinion theory (Romer, 1993, Hong and Stein, 2003)
  - volatility feedback effect (French et al., 1987, Campbell and Hentschel, 1992)

Background

- The managerial incentives to conceal bad news (Hutton et al., 2009, Jin and Myers, 2006)



Background

Increase crash risk

Opaque financial reporting, (Hutton et al., 2009)

Corporate tax avoidance (Kim et al., 2011b)

Executive equity incentives (Kim et al., 2011a)

Corporate social responsibility performance (Kim et al., 2014)

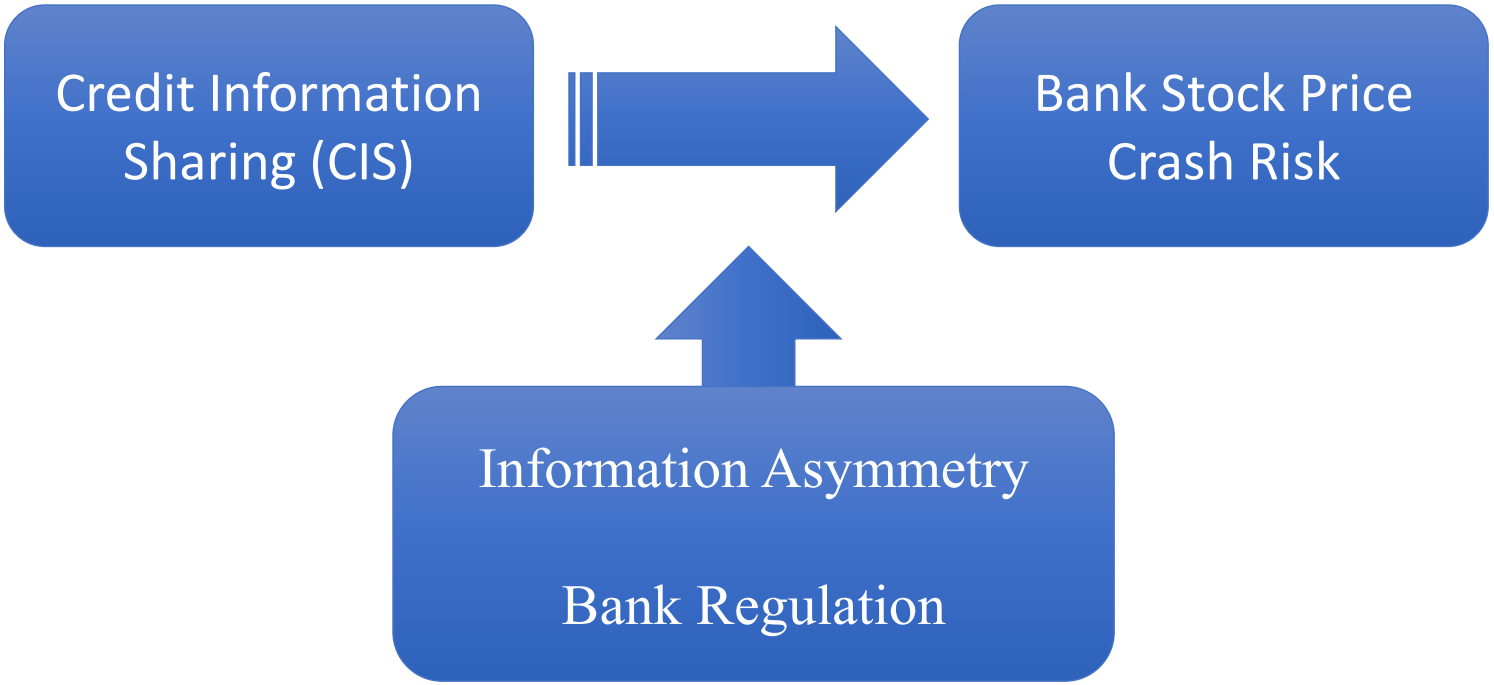
Accounting conservatism (Kim and Zhang, 2015)

Decrease crash risk

Background

- It is the ability of banks to make loans that make them opaque. The quality of loan is not readily observable and can even be hidden for a long period of time (Morgan 2002; Flannery et al. 2004; Levine 2004).
  - Because of the imperfect information in the lending process, a loan officer has a significant amount of discretion in terms of the lending decision. Such amount of discretion creates a room for bribery and leads to corruption in lending (Barth et al. 2009).
  - Due to career concern, a loan officer tends to suppress unfavorable information about their assigned borrowers' repayment prospects (Hertzberg et al. 2010).
  - reflect poorly on how she has performed
- Unprofitable projects are kept alive and poor performance accumulates over time (Bleck & Liu 2007).
- Eventually, the accumulation is no longer possible. The release of all negative information leads to share-price collapse.

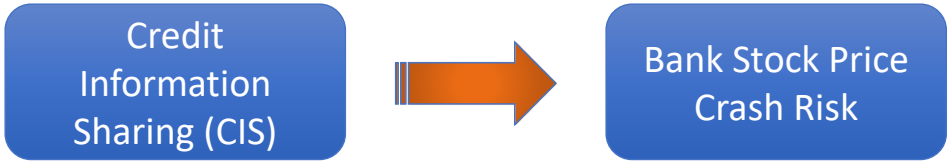
Research Objectives



Hypotheses Development

- Credit information sharing may discourage loan officers to withhold negative information:
  - According to Barth et al. (2009), CIS facilitates the monitoring of loan officers and helps decrease bank-lending corruption by reducing the degree of discretion that can be exercised in evaluating loan applicants.
  - Information shared on a credit bureau/registry can be used to validate the internal risk ratings assigned by loan officers and ensure that the information content from reports issued by loan officers are unbiased.
  - CIS also Improves comparability. Investors gain a better understanding of a performance of a firm and also obtain some of the bad news by inferring the performance and/or disclosure of the firm's comparable peers (Kim et al. 2016).

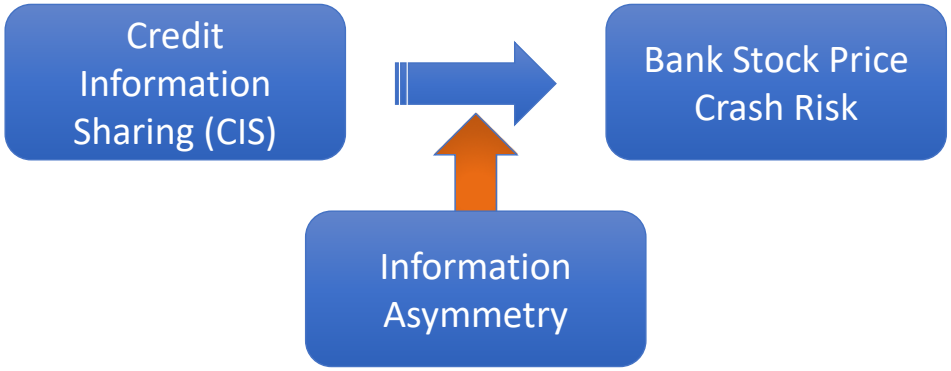
H1: Credit information sharing ***reduces*** bank-specific stock price crash risk



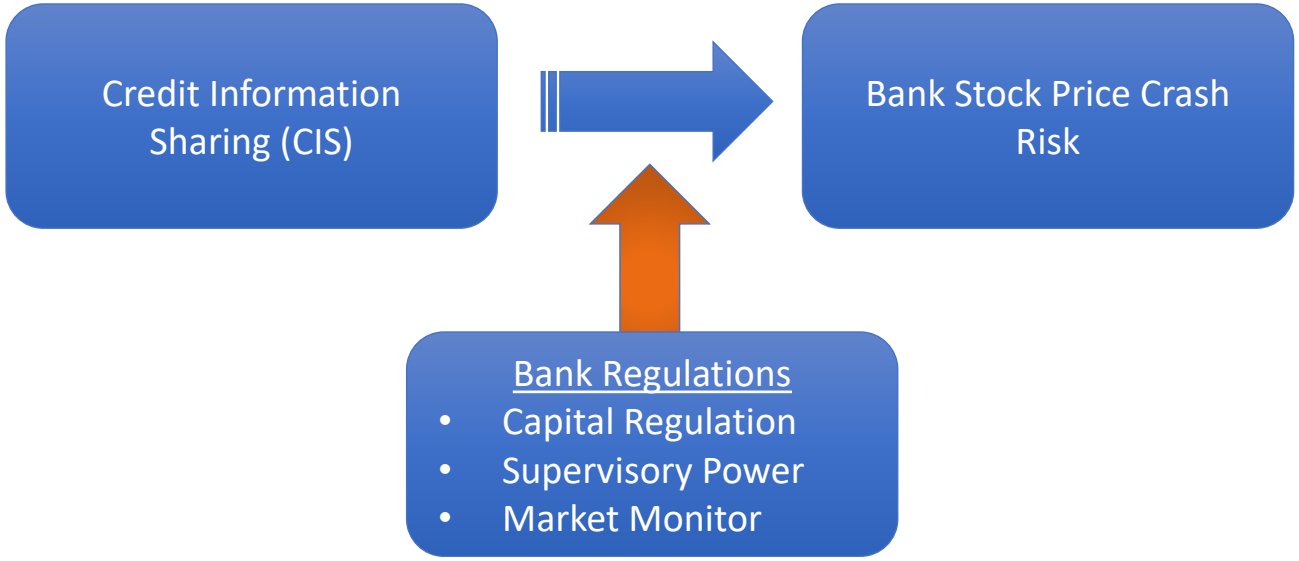
Hypotheses Development

- In an environment with less transparent information environment, information sharing should be particularly more useful for banks because more information is accessible to both investors and managers.
- In such environment, loan officers have less ability to conceal negative information about their assigned borrowers.

H2: The impact of credit information sharing on stock price crash risk is expected to be ***less pronounced*** when the information environment is **more** transparent.



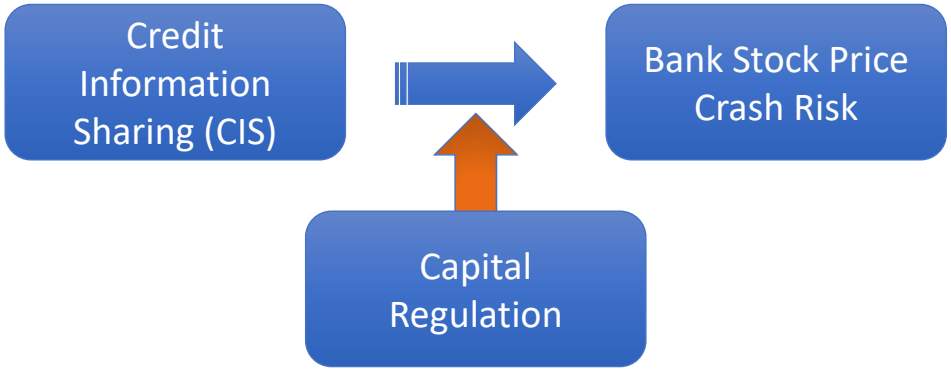
Hypotheses Development



Hypotheses Development

- Capital requirements reduce gambling incentives and promote prudent behavior by putting bank equity at risk (Hellmann et al. 2000).
- If banks are not required to put a substantial amount of their equity at risk, they may not be cautious in lending and might take an excessive risk like concealing bad news for an extended period.

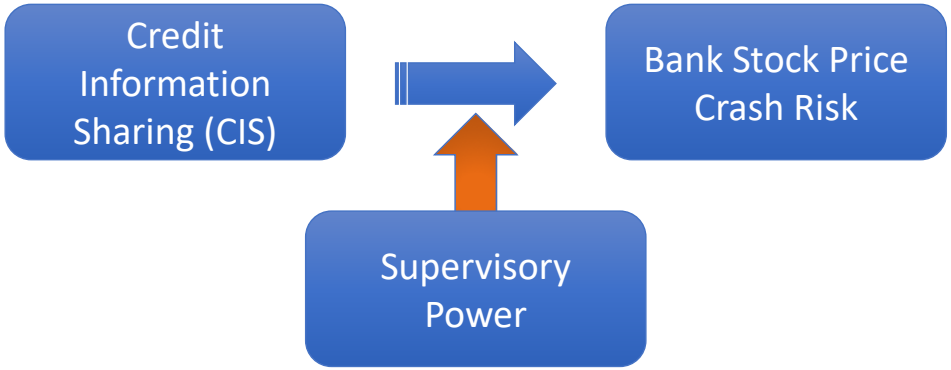
H3-A: The impact of credit information sharing on stock price crash risk is expected to be ***more pronounced*** in a banking regulatory environment with ***less*** stringent capital requirement.



Hypotheses Development

- In general, bank supervisors have responsibility and expertise to overcome market imperfection.
- In this regards, supervisors need significant powers to prevent banks from engaging in undesirable activities and from taking excessive risks, especially in light of the growing complexity of banking activities (Stigler 1971; Beck et al. 2006b).

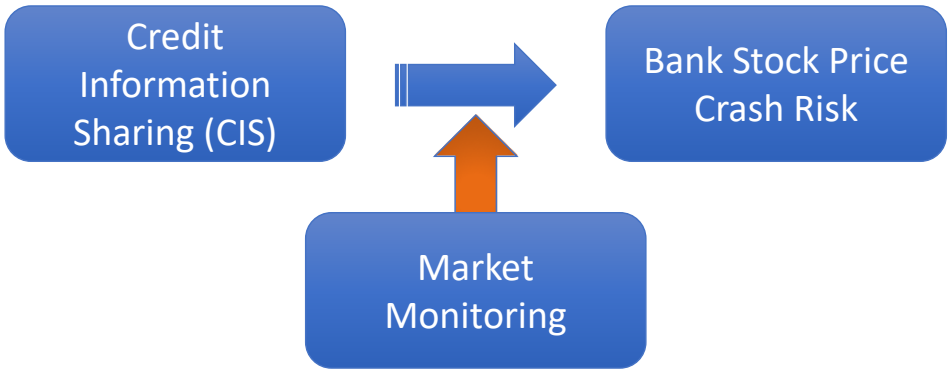
H3-B: The impact of credit information sharing on stock price crash risk is expected to be ***more pronounced*** in a banking regulatory environment with **low** supervisory power.



Hypotheses Development

- Market monitoring refers to empowering private agents/investors to monitor banks by increasing information disclosure to officials, the public and specialized entities such as rating agencies and auditors.
- Policies should thus focus on inducing banks to disclose more and accurate information to the public so that the private agents/investors have enough ability to overcome information barriers. Consequently, they can exert effective monitoring and governance over banks.

H3-C: The impact of credit information sharing on stock price crash risk is expected to be ***more pronounced*** in a banking regulatory environment with **low** degree of market monitoring



Data & Sample

- Database:
  - Datastream (Thomson Reuters)
  - BankScope (BvD)
  - World Bank's Doing Business
  - World Bank's World Development Indicators (WDI)
  - World Bank's Bank Regulation & Supervision Survey
- Sample:
  - 1,402 listed banks in 55 countries
  - 7,562 bank-year observation
  - Period: 2005-2013 (Annual)
- Control variables:
  - Bank-specific controls (market value of equity, market-to-book-ratio, leverage, ROA, capital ratio, deposits, Δstock volume, stock volatility, average past return)
  - Country-level controls (GDP growth, stock market capitalization, GDPpc)

Variable Measurement

- The firm-specific weekly return for firm i in week t, ($w_{i,t}$) as followed:

$$W_{i,t} = \log(1 + \varepsilon_{i,t})$$

<u>The expanded market model</u>

$$\begin{aligned} r_{i,t} & \\ &= \alpha_i + \beta_{1,i} r_{m,j,t} + \beta_{2,i}\left[r_{U.S.,t} + EX_{j,t}\right] + \beta_{3,i} r_{m,j,t-1} + \beta_{4,i}\left[r_{U.S.,t-1} + EX_{j,t-1}\right] \\ &+ \beta_{5,i} r_{m,j,t-2} + \beta_{6,i}\left[r_{U.S.,t-2} + EX_{j,t-2}\right] + \beta_{7,i} r_{m,j,t+1} + \beta_{8,i}\left[r_{U.S.,t+1} + EX_{j,t+1}\right] \\ &+ \beta_{9,i} r_{m,j,t+2} + \beta_{10,i}\left[r_{U.S.,t+2} + EX_{j,t+2}\right] + \varepsilon_{i,t} \end{aligned} \quad (1)$$

Where $r_{i,t}$ is the stock return for firm i in week t, $r_{m,j,t}$ is the local market return for country j in week t, $r_{U.S.,t}$ is the U.S. market return in week t (a global market return proxy), and $EX_{j,t}$ is the change in exchange rate for the currency of country j against the U.S. dollar in week t.

Variable Measurement

- **Crash Risk:**
  - Negative conditional skewness ($NCSKEW$)

$$NCSKEW_{i,t} = -\left[n(n-1)^{3/2}\sum W_{i,t}^{3}\right] \Big/ \left[(n-1)(n-2)\left(\sum W_{i,t}^{2}\right)^{3/2}\right]$$

Where $W_{i,t}$ is firm-specific weekly return , and n is the number of weekly return observation of year t.

A higher value indicates higher crash risk

Variable Measurement

- **Crash Risk:**
  - Down-to-up volatility ($DUVOL$)

$$DUVOL_{i,t} = log\left[(n_u - 1)\sum_{DOWN} W_{i_d,t}^2 \Big/ (n_d - 1)\sum_{UP} W_{i_u,t}^2\right]$$

The group with "Down" weeks consists of the returns that are below the yearly mean. The group with "Up" weeks consists of the returns that are above the yearly mean. Where $W_{i_d,t}$ and $W_{i_u,t}$ are the firm-specific returns for down- and up-weeks, respectively; n_u and n_d are the numbers of up- and down- weeks in year t, respectively.

A higher value is associated with higher level of crash risk.

Variable Measurement

	Variable	Definition	Source
Explanatory Variables (Country-level)	Depth of Credit Information Sharing Index (DEPTH)	The index measures the scope, accessibility, and quality of credit information available through information sharing institutions (ranging from 0 – 6)	World Bank's Doing Business
Proxies for Credit Information Sharing	Private Credit Bureau Coverage (PRIV)	The number of individuals and firms listed by a private credit bureau with information on repayment history or unpaid debts scaled by the adult population (%)	
	Public Credit Registry Coverage (PUB)	The number of individuals and firms listed in a public credit registry with information on repayment history or unpaid debts scaled by adult population (%)	
Proxies for Information Environment	IFRS Adoption (IFRS)	A dummy 1 for a country with mandatory IFRS adoption and 0 otherwise	"IFRS foundation and IASB", Deloitte
	Business Extent of Disclosure Index (BDI)	The extent to which investors are protected through disclosure of ownership and financial information (ranging from 0 – 10)	World Bank's Doing Business

Variable Measurement

Variable		Definition	Source
Explanatory Variables (Country-level)	Low Capital Stringency Index (LOW_CAPITAL_STR)	• CAPITAL_STR is an index measuring the extent of both initial and overall capital stringency (ranging from 0 – 10) • A dummy value of 1 for a country with a value of CAPITAL_STR lower than the sample 50 th percentile and zero otherwise.	World Bank's Bank Regulation and Supervision Survey Database
	Low Official Supervisory Power index (LOW_SUPER_POW)	• SUPER_POW is an index measuring the power of officers in supervising and monitoring banks .(ranging from 0 – 14) • We assign a dummy variable equal to 1 for a country with a value of SUPER_POW lower than the sample 50 th percentile and zero otherwise.	
	Low Private Monitoring Index (LOW_MONITOR)	• MONITOR is an index measuring the degree of which regulations empower the private investors to monitor and exert effective governance over banks (ranging from 0-10) • We assign a dummy variable equal to 1 for a country with a value of MONITOR lower than the sample 50 th percentile and zero zero otherwise.	

Bank Regulationst

Methodology

Estimation Model:

Panel-Data Model with bank-fixed effects plus time dummies

CRASH: crash risk measure (NCSKEW)
CIS: credit information sharing proxies (DEPTH, PRIV, PUB)
ASYM: Information environment proxies (IFRS, BDI)
X: Firm-specific controls
Y: Country-specific controls

-

$$CRASH_{i,t} = \beta_0 + \beta_1 CIS_{i,t-1} + \sum_{k=2}^{10} \beta_k (X_{i,t-1}^k) + \sum_{m=11}^{13} \beta_m (Y_{i,t-1}^m) + \lambda_t + \alpha_i + \varepsilon_{i,t}$$

+

$$CRASH_{i,t} = \beta_0 + \beta_1 CIS_{i,t-1} + \beta_2 ASYM_{i,t-1} + \beta_3 ASYM_{i,t-1} * CIS_{i,t-1} + \sum_{k=4}^{12} \beta_k (X_{i,t-1}^k) + \sum_{m=13}^{15} \beta_m (Y_{i,t-1}^m) + \lambda_t + \alpha_i + \varepsilon_{i,t}$$

Methodology (Con't)

$$CRASH_{i,t} = \beta_0 + \beta_1 CIS_{i,t-1} + \beta_2 LOW_CAPITAL_STR_{i,t-1} + \beta_3 LOW_CAPITAL_STR_{i,t-1} * CIS_{i,t-1} + \sum_{k=4}^{12} \beta_k (X_{i,t-1}^k) + \sum_{m=13}^{15} \beta_m (Y_{i,t-1}^m) + \lambda_t + \alpha_i + \varepsilon_{i,t}$$

$$CRASH_{i,t} = \beta_0 + \beta_1 CIS_{i,t-1} + \beta_2 LOW_SUPER_POW_{i,t-1} + \beta_3 LOW_SUPER_POW_{i,t-1} * CIS_{i,t-1} + \sum_{k=4}^{12} \beta_k (X_{i,t-1}^k) + \sum_{m=13}^{15} \beta_m (Y_{i,t-1}^m) + \lambda_t + \alpha_i + \varepsilon_{i,t}$$

$$CRASH_{i,t} = \beta_0 + \beta_1 CIS_{i,t-1} + \beta_2 LOW_MONITOR_{i,t-1} + \beta_3 LOW_MONITOR_{i,t-1} * CIS_{i,t-1} + \sum_{k=4}^{12} \beta_k (X_{i,t-1}^k) + \sum_{m=13}^{15} \beta_m (Y_{i,t-1}^m) + \lambda_t + \alpha_i + \varepsilon_{i,t}$$

LOW_CAPITAL: a dummy for capital stringency index
LOW_SUPER_POW: a dummy for supervisory power index
LOW_MONITOR: a dummy for private monitoring index

Empirical Results

$$CRASH_{i,t} = \beta_0 + \beta_1 CIS_{i,t-1} + \sum_{k=2}^{10} \beta_k (X_{i,t-1}^k) + \sum_{m=11}^{13} \beta_m (Y_{i,t-1}^m) + \lambda_t + \alpha_i + \varepsilon_{i,t}$$

Variable	NCSKEW		
	(1)	(2)	(3)
DEPTH	-0.009		
	(-0.30)		
PRIV		0.028	
		(0.32)	
PUB			-0.462***
			(-2.72)

$$CRASH_{i,t} = \beta_0 + \beta_1 CIS_{i,t-1} + \beta_2 ASYM_{i,t-1} + \beta_3 ASYM_{i,t-1} * CIS_{i,t-1} + \sum_{k=4}^{12} \beta_k (X_{i,t-1}^k) + \sum_{m=13}^{15} \beta_m (Y_{i,t-1}^m) + \lambda_t + \alpha_i + \varepsilon_{i,t}$$

Variable	NCSKEW		NCSKEW	
	(1)	(2)	(3)	(4)
PUB	-0.464***	-0.870***	-0.461***	-0.413*
	(-2.69)	(-6.91)	(-2.72)	(-1.70)
IFRS	-0.022*	-0.052		
	(-1.77)	(-1.17)		
*IFRS * PUB*		0.510*		
		(1.91)		
LOW_BDI			-0.012	-0.002
			(-0.17)	(-0.03)
*LOW_BDI * PUB*				-0.121
				(-0.40)

Empirical Results (Con't)

Variable	*NCSKEW*		*NCSKEW*		*NCSKEW*	
	(1)	(2)	(3)	(4)	(5)	(6)
PUB	-0.481***	-0.454**	-0.463**	-0.436**	-0.465**	-0.465**
	(-2.78)	(-2.53)	(-2.57)	(-2.40)	(-2.63)	(-2.63)
LOW_CAPITAL_STR	0.226*	-0.09				
	(1.89)	(-0.49)				
LOW_CAPITAL_STR * PUB		-3.389***				
		(-3.55)				
LOW_SUPER_POW			0.016	0.138		
			(0.13)	(1.17)		
LOW_SUPER_POW * PUB				-5.124***		
				(-3.69)		
LOW_MONITOR					0.041*	0.074
					(1.71)	0.85
LOW_MONITOR * PUB						-4.146***
						(-8.08)

Robustness Tests

- Alternative Measures of Crash Risk
  - Down-to-Up Volatility
- Additional Controls
  - Series of macro institutional indexes from WGI
  - Conditional firm-year conservatism measure (C_SCORE)
- Subsample Analysis
  - More V.S. Less transparent information environment
  - Strong V.S. Weak Banking Regulatory Environment
- Exclude USA banks
- Instrumental Variable (IV) approach
  - Legal origins, ethnic fractionalization, and latitude
- Additional Tests
  - Deposit insurance mitigates the relationship between crash risk and credit information sharing
  - Activity restriction has no notable impact the relationship between crash risk and credit information sharing

Conclusion

- More credit information sharing through public credit registries is negatively associated with stock price crash risk.
- Our findings are consistent with the view that forcing banks to share borrower information sharing may discourage loan officers to withhold negative information. Consequently, negative information is less likely to be accumulated and the probability of crash decreases.
- The negative effect of credit information sharing on crash risk is attenuated in an environment with higher information transparency.
- The impact of information sharing on crash risk is more pronounced with:
  - Less stringent capital requirements
  - Less supervisory power
  - Less emphasis on market-based monitoring

Policy Implications

- Policymakers may need to focus on the establishment and improvement of public credit registries in order to prevent stock price crash in banking sectors.
- When policymakers decide to uphold information sharing among banks, it is necessary to take into consideration the transparency of the information environment.
- Where banking regulatory environment is weak or less stringent, policies should aim to the development of public credit registries to reduce crash risk.

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THANK YOU

Depth of Credit Information Sharing

- An index that measures the scope and contents of credit information that being shared. It ranges from zero to six. The value of zero indicates that there is no public credit registry or private credit bureau operating in a country. The value of one is added to the index with each of the following characteristics:
  - Both positive and negative information are distributed.
  - Data on households and firms are distributed.
  - Data from retailers, trade creditors, and/or utility companies as well as financial institutions are distributed.
  - More than 2 years of data are available.
  - Data are collected and distributed on loans with value below 1% of income per capita.
  - Laws give right to borrowers to inspect their own data.

Asset Diversification Index

- an index measuring whether there are explicit, verifiable, and quantifiable guidelines for asset diversification (e.g., if banks are required to have some minimum diversification of loans among sectors, or if there are sectoral concentration limits); and whether banks can make loans aboard. Higher values of the index indicate more bank asset diversification.

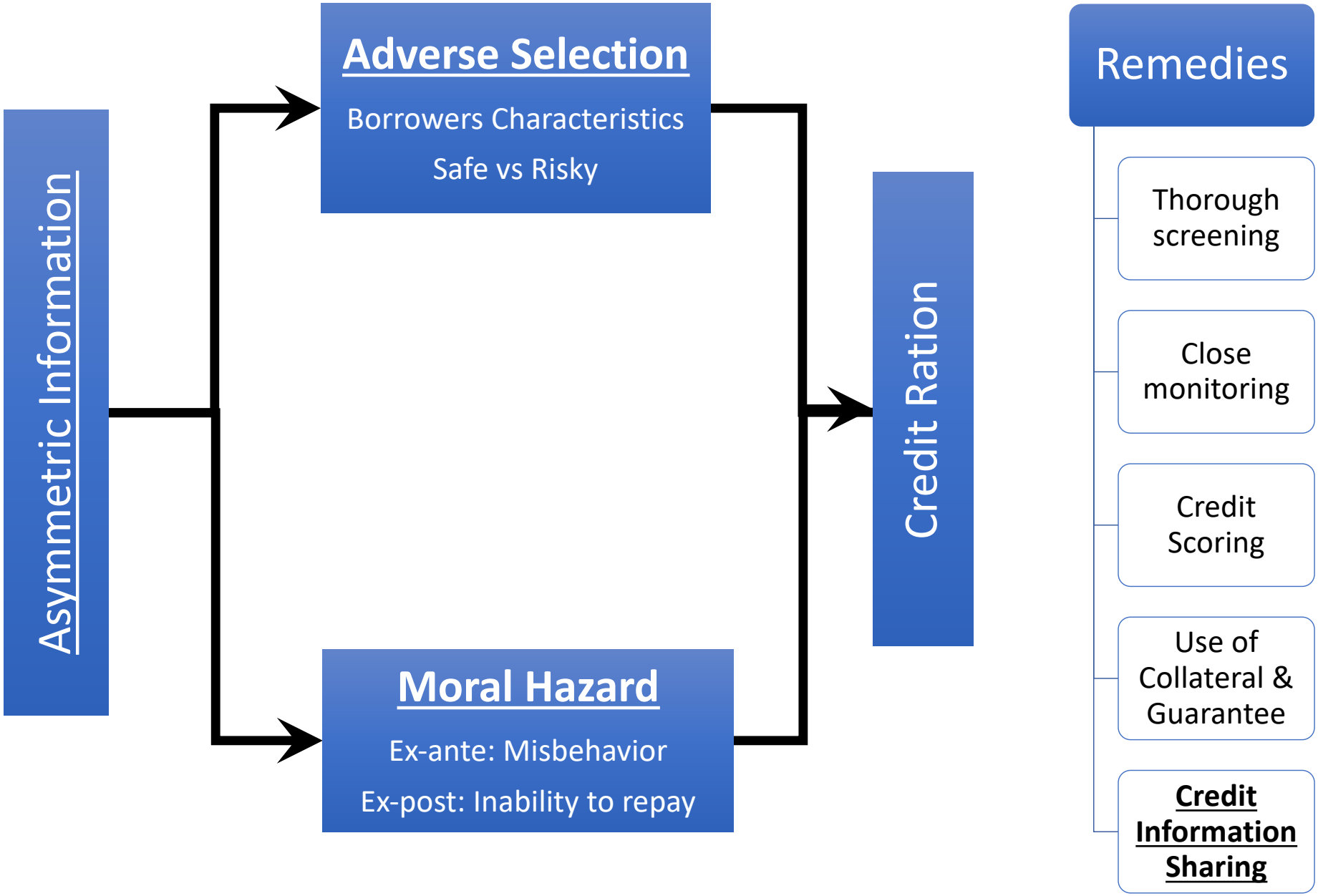
Suggestions for Future Research

- In this study, we use depth of credit information sharing index (DEPTH) is a good proxy measuring the scope, coverage, and accessibility of credit information available through information sharing agencies in each country. However, this depth index for each country is publicly available only in a single index aggregating six individual features. Thus, it would be helpful if data underlying the depth of credit information index becomes publicly available so that this information can easily be used for research.
- Examine whether the effect of credit information sharing depends on the extensiveness of information sharing as proxied by the number of private credit bureaus a bank is partnering within a given year.
- The impact of credit information sharing may carry over to macro-level concerning economic development and poverty reduction. Unequal access to finance has long been recognized as a critical mechanism for generating persistent income inequality and slower economic growth (Beck et al. 2007).

Contributions

- Add to the literature of CIS by identifying the new economic consequence of credit information sharing in banking sectors
- Add to the literature of crash risk by identifying a new factor affecting crash risk.
- Emphasize the role of bank regulation and supervisions as an external governance over banks' behaviors.

Background


Asymmetric Information
Adverse Selection
Borrowers Characteristics
Safe vs Risky
Moral Hazard
Ex-ante: Misbehavior
Ex-post: Inability to repay
Credit Ration
Remedies
Thorough screening
Close monitoring
Credit Scoring
Use of Collateral & Guarantee
Credit Information Sharing